Guidance and staff card

At check-out, record E for … against the pupil's target.

At check-out, record E for explaining how a stem cell divides and how some descendants specialise against the pupil's target.

Prepared 3 November 2026

Invite a response. Give thinking time. The learner may pass. Offer private answers and the usual response mode.

Return to learning: agree a quiet pause, offer the regulation words beside the colours, then return to one labelled arrow or one process choice. Use the individual plan.

Use soft instrumental music for transitions if the class plan says so.

Code	Meaning
VF	Verbal feedback given here
WS	Worked with support
I	Independent
NS + …	Next step one specific thing
E	Evidence captured on the evidence app
R	Responded loop closed after the pupil responds and an adult engages
//	Self-edit point
?	Read this back to me

Use written feedback for extended writing when it adds value. An edit is in the same colour, on the same page. Buttons and saved responses do not award feedback codes.

Codes follow the verified feedback policy, Issue 1. The curriculum summary lists the first six codes; the feedback policy adds the edit and read-back codes and supplies the loop-closure meaning.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil’s improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt

Teacher and cover plan

At check-out, record E for explaining how a stem cell divides and how some descendants specialise against the pupil's target.

I can explain how a stem cell divides and how some of its descendants become specialised for a job.

Prior content: making more cells, specialised roles, animal tissue and plant tips. Start from responses; do not assume attendance.

Stage	Minutes	Teaching and check
Today I can	1	Which change concerns cell number, and which concerns a specialised job?
Arrival	5	Use Q1 to check picture access before making a science judgement. Give thinking time, invite a response and accept a pass.
Starter	2	Collect a separate reason from each learner. If number and role are confused, use two plain counters before adding a cell-type label.
I do	4	Model one complete explanation. Ask which word names the change in number, then which names the change in role.
We do	5	Invite each learner to choose and explain before checking. Change source after a correct example. Fade the cue for a fresh pair.
I do	3	Point to the root-tip region. Link each process to what it changes. Distinguish an individual cell from the root as a whole.
We do	4	Hear each reasoning chain. If only order is secure, ask a fresh outcome-to-process question. Do not accept a correct click as an explanation.
You do	10	Give access support without dictating the science. Hear each learner’s own feature-function link. Mark I, S or C beside the actual task.
Review and improve	3	Respond to one real piece of work. Wait for the pupil’s revision or reason, then engage with it. A button does not close the feedback loop.
Exit ticket	3	Hear the two checks once in a valid mode. Use the least secure dependency to choose the next teaching move. Access support and content cues are recorded separately.

Total: 40 minutes. Adapt pace to access and readiness. Preserve one model, an independent response, review and the exit.

Week arc: Discover: growth and stem cells. Use: how stem cells become specialised cells. Master: growth, differentiation and stem-cell evidence.

Equipment: large prepared process cards, labelled desk references, writing materials and an offline display if available. No biological samples are required.

Stage timings

Stage	Minutes
Today I can	1
Arrival	5
Starter	2
I do	4
We do	5
I do	3
We do	4
You do	10
Review and improve	3
Exit ticket	3
Tools & print	0

Total: 40 minutes

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil's improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Exit pupil voice
What helped you most today, or what would you change next time?
Accept a pointed, spoken, signed, drawn, written or chosen response. A pupil may pass.

Made by Matt

P17 · SLT planning and evaluation

Planned adaptations

Need	Planned adaptation	Where it is used
Cognition and learning	One outcome-to-process link, then fade the cue	Arrival and first model
Communication and interaction	Thinking time, pair rehearsal, point, sign or exact scribe	Every question and model checkpoint
Regulation	Private adjustment, quiet pause, one return step	Arrival and You do
Sensory and physical	Static drawings, enlarged cards, adult-directed handling	Both working models
Language access	Pre-teach word and picture; permit planning in any language	Word help and Reference A

If behind

Keep division versus differentiation, one source-range decision and a feature-function link. Shorten the comparison task, then retain review and two exit checks. Give Reference B beside the shortened task.

Where catch-up is embedded

Arrival Q1 uses picture labels. The two I do models re-teach the dependencies. Reference A and B stay available. The catch-up sheet replaces an access step only; it is never an extra routine task.

Staff evaluation

Prompt	Record after teaching
Delivered as planned	Yes / Partly / No; actual change and reason
Criteria met	Criterion, response and support actually used
Adaptation and effect	What changed and what fresh response showed
Coverage impact	What remains and the next teaching move
Evidence	Location on the evidence app and lesson code

Made by Matt

P18 · Individual target opportunities

Choose opportunities from the learner’s existing plan. Do not pre-fill pupil codes, targets or outcomes.

Opportunity	Lesson evidence	Adult action
Cognition and learning	Select division for more cells, then answer a fresh example	Fade one content cue
Communication	Explain one source-target decision in a chosen mode	Wait, listen, then check the meaning
Regulation	Request an adjustment and return to one arrow	Follow the individual plan and check access
Sensory or physical access	Direct a model choice or card placement	Record access separately from science support
Independence	Give a fresh feature-function link	Mark the actual support at that task

Keep one target-matched record on the evidence app where it adds value. Task completion alone does not demonstrate a personal target.

Made by Matt

Adult deployment and responsive feedback

Before teaching, agree who leads the model and who checks access. With two adults, one supports an access barrier while the other hears the remaining learner; swap roles as needed. With one adult, use a prepared independent reference task while hearing each learner separately.

Wait → gesture to the relevant part → rephrase → give a word bank → reduce to a limited choice. Teach one step, then check a fresh example. Avoid telling the final answer.

At the first checkpoint, hear the source-range reason separately from the process reason. During You do, the teacher checks a feature-function link and the plant comparison. Mark actual I · S · C beside each task.

Give one precise next step. Hear the learner’s revision or explanation and use it to change the next prompt. Access support is not automatically a science content cue.

Made by Matt

First lesson back · once-only read

Before the lesson · scan the books

Scan for five minutes. Find where teaching stopped and what responses were actually made. Look for access barriers and useful science evidence.

Existing NS + is history

Do not chase, back-mark or retrospectively close earlier feedback. Begin fresh feedback from this lesson.

Ask the cover teacher and TAs

Ask what was taught, which models were used, what changed for individuals and which responses were heard.

Check the roll

Compare attendance with the available evidence. New arrivals start from what they show here; absence creates no automatic deficit label.

Arrival read-off · decide by about minute eight

Q1 checks picture access. Use Q2 for number, Q3 for inherited information and Q4 for plant growth. If number and job are confused, use the first model. If secure, fade the cue and use a new source-target example.

After last lesson · one read, once

Read each exit against its criterion and the support used. Choose the next route from the least secure dependency. Teach next; do not reopen historical feedback.

Made by Matt

Preparation and practical safety

This is a model lesson. It uses prepared cards and drawings. It does not involve culturing cells, collecting tissue, handling blood, treatment decisions or a claimed core practical.

Check large cards and the desk area. Keep loose cards on a mat and pick up fallen items. Provide seated or adult-directed handling where needed. Follow the local risk assessment and individual plans.

Test the offline display, keyboard controls, print choices and reset actions. Keep a static reference available. Use each learner’s usual access equipment.

Destination link: explaining cell structure and interpreting scientific claims supports further biology study and work in laboratory support and biomedical science.

Made by Matt

Teacher answers and topic guidance

Accept scientifically equivalent wording and the usual response mode. Keep answers hidden during independent attempts.

Starter

More cells supports division. It does not alone establish differentiation or a specialised role.

Models

The source-range checkpoint accepts embryonic → nerve, muscle or blood; adult blood-forming → blood; meristem → root hair. Differentiation gives a specialised role. Reject animal-to-plant routes and the claim that all adult stem cells form all cell types.

Plant chain: division → elongation → differentiation in this simplified route. The claim about every cell being a stem cell is unsupported by length alone.

Arrival Supported

1. A. Read the label beside the round cell.
2. More cells.
3. The nucleus.
4. Root tip.

Arrival Independent

1. A.
2. More cells with matching inherited information.
3. Division makes more cells; differentiation gives a specialised role.
4. In the root-tip meristem.

Arrival Aspire

1. A, stem cell.
2. Inherited DNA is copied; two daughter cells receive matching inherited information.
3. Division is shown. Differentiation explains specialised roles.
4. The new cells can elongate, increasing their length.

Exit Supported

1. Differentiation.
2. More cells.

Exit Independent

1. Differentiation.
2. Elongation increases cell length; or differentiation gives a specialised role.

Exit Aspire

1. Cell number shows division; a role needs evidence of differentiation and features suited to a function.
2. Elongation or division can contribute; length alone does not identify every cell type or prove stem-cell range.

Topic guidance

Keep stem cell as a cell with a capacity, not a name for every newly made cell. Adult stem-cell types have limited ranges. Embryonic stem cells can form many body-cell types; avoid the claim that all stem cells can form all cell types.

Mitosis is nuclear division. Cell splitting produces separate cells. The teaching drawings simplify the full cell cycle. Mature specialised cells can differ greatly in form; do not claim every specialised cell contains a nucleus.

A feature-function link is an explanation. A schematic drawing supplies a model, not an experimental observation or proof of clinical success.

Mark scheme Supported

Award each distinct point once. These are question tariffs, not participation scores. Use actual responses and record content support honestly.

1. Finish: Division makes more ___. Differentiation gives a specialised ___. [2]

cells; job.

Maximum 2 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

2. Use Reference A. Choose nerve or muscle for a cell with a long extension. Say how that feature helps. [3]

Nerve cell; long extension; carries signals over a distance.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

3. Choose the possible route: an adult blood-forming stem cell makes blood cells or every kind of body cell. Give a reason. [3]

Blood cells; adult stem-cell types have a limited range; the model does not show every body-cell type.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

4. Put the plant cards in order: differentiate, divide, elongate. Name the region where new cells are made. [4]

Divide, elongate, differentiate; meristem.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

5. Compare the skin and root models. Finish: Both make more cells by ___. The root cells also ___. A cell gains a job by ___. [3]

division; elongate; differentiation.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

Made by Matt

Mark scheme Independent

Award each distinct point once. These are question tariffs, not participation scores. Use actual responses and record content support honestly.

1. Explain the difference between division and differentiation. [2]

Division makes more cells; differentiation makes a cell specialised for a particular job.

Maximum 2 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

2. Use a labelled nerve-cell drawing to link one feature to its function. [3]

Labelled long extension; signal transmission; link to carrying a signal over a distance.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

3. Compare the range of an embryonic stem cell with that of an adult blood-forming stem cell. State one limit of the model. [3]

Embryonic stem cells can give rise to many body-cell types; adult blood-forming stem cells produce a more limited range; model cannot prove successful treatment or show all stages.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

4. Draw and label a simplified growth route at a root tip. Include division, elongation, differentiation and meristem. [4]

Meristem named; division adds cells; elongation lengthens cells; differentiation gives specialised roles.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

5. Compare growth and repair in animal skin with growth in a root. Give one shared process and one difference, with an explanation. [3]

Both include division; both may include differentiation; root-tip meristem followed by elongation is a relevant plant difference. Credit an equivalent clear three-point comparison.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

Mark scheme Aspire

Award each distinct point once. These are question tariffs, not participation scores. Use actual responses and record content support honestly.

1. A model shows two new cells, both labelled specialised. Explain why cell number alone cannot justify those labels. Add the missing process. [3]

Number supports division; number alone does not show a specialised feature or role; add differentiation with a feature-function link.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

2. Compare a nerve cell and muscle cell. Link one feature of each to its job, then state what a drawing cannot establish. [4]

Nerve extension; signal transmission; contractile muscle structures linked to contraction; drawing alone cannot establish measured function.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

3. Evaluate: Every adult stem cell can become any body cell. Correct the scope and use the blood-forming example. [4]

Reject universal claim; adult stem-cell types have limited ranges; blood-forming stem cells make blood cells; embryonic stem cells have a wider range of body-cell types.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

4. Compare the skin and root routes. Explain two shared ideas and one plant difference. Give one limit of a simple three-step sequence. [5]

Division shared; differentiation shared; root-tip meristem; elongation contributes to length; sequence is simplified and processes can overlap across cells.

Maximum 5 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

5. Repair: A longer root proves every new cell is a stem cell. State what growth could show, what it does not show and a relevant extra check. [4]

Length indicates growth; could involve division and elongation; does not establish every cell is a stem cell; inspect cell features or location using an appropriate prepared reference, without claiming that appearance alone proves potency.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

Made by Matt

Mark scheme Choice

Award each distinct point once. These are question tariffs, not participation scores. Use actual responses and record content support honestly.

1. Choose division or differentiation for makes more cells. Choose again for gives a job. Add one short reason. [2]

division; differentiation, with a matching reason.

Maximum 2 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

2. Choose nerve or muscle for carrying signals. Choose long extension or no structure as a useful feature. Explain one link. [3]

nerve; long extension; carries signals over a distance.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

3. Choose blood cells or all body cells for the range shown from an adult blood-forming stem cell. Say one limit. [3]

blood cells; limited adult range; does not prove treatment success.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

4. Select a plant sequence: divide then elongate then differentiate, or differentiate then no further change. Name root-tip meristem or leaf middle as the dividing region. [4]

divide, elongate, differentiate; root-tip meristem.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

5. Select both, animal only or plant only for division. Select elongation or signal transmission as the length change in the root model. Give one comparison sentence. [3]

both; elongation; a clear comparison linking the process to its effect.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

Made by Matt

Science staff guidance

Invite a response, give thinking time and allow a pass. Do not ask a learner to read difficult text aloud without agreement. Accept pointing, sign, drawing, exact scribing and adult-directed handling.

State which barrier you saw: reading, vocabulary, processing, regulation, writing or science. Check access before judging a non-response. Avoid prompting through every question.

Secure division but insecure differentiation: use the same cell count with a new feature label. Secure job but insecure source range: compare adult blood-forming with embryonic source. Secure animal example but insecure plant route: trace number, length and job on Reference B.

At Review, engage with one real revision. At Exit, read the response once and choose the next teaching move. Keep the usual-mode response as the evidence rather than adding a duplicate form.

Made by Matt

TA Brief

At check-out, record E for explaining how a stem cell divides and how some descendants specialise against the pupil's target.

I can explain how a stem cell divides and how some of its descendants become specialised for a job.

Key checks: division changes number; differentiation gives a specialised role; stem-cell types differ in their range; a root can grow through division and elongation.

At Arrival, check picture access before giving a science cue. At the source-target model, hear a reason before checking. At You do, support access and hear a fresh answer. Use the actual staffing plan.

Wait, point, rephrase, then offer a word bank or limited choice. Record actual I · S · C beside the task. Return to one labelled arrow after a quiet pause.

Answers: starter no; first checkpoint uses differentiation; adult blood-forming source makes blood cells; plant chain division, elongation, differentiation. Use Teacher answers for the full scheme.

Report the barrier and the response, not a general label about ability. A model choice is not proof of understanding without a reason.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
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